MODULE 42 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

CI/CD Pipeline Design

Build once, promote the same artefact through every account, and let the pipeline itself decide when a release may enter or leave a stage.

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One pipeline in a **tooling account** builds an artefact **once**, then assumes a role in each target account to deploy that same artefact.

The developer tools

SERVICE	JOB	EXAM NOTE
CodeCommit	Managed Git	Back to full GA for new customers, Nov 2025
CodeConnections	Link GitHub, GitLab, Bitbucket	Source and triggers
CodeBuild	Build and test	Also runs pipeline rules
CodeDeploy	Deploy to EC2, Lambda, ECS	Traffic shifting, alarm rollback
CodePipeline V2	Orchestrate stages	Triggers, variables, conditions

ON THE EXAM

CodeCommit is open to new customers again. Older material saying otherwise is out of date.

Cross-account deployment

01

Tooling account

Pipeline and artefact bucket

02

Customer managed key

Key policy allows target accounts

03

Assume deploy role

Role in each target account

04

Deploy

CloudFormation, CodeDeploy or ECS

ON THE EXAM

The AWS managed S3 key cannot be shared across accounts. Use a customer managed KMS key.

CodePipeline V2 controls

FEATURE	WHAT IT DOES
Trigger filters	Start on branch, file path, tag or pull request
Variables	Pass values at start and between actions
Execution mode	Superseded, queued or parallel
Entry condition	Gate a stage before it starts
On-success, on-failure	Roll back, fail, retry or skip
Rules	CloudWatch alarm, deploy window, Lambda, commands

ON THE EXAM

Stage rollback is not available in parallel execution mode.

Build once or build per environment?

BUILD ONCE, PROMOTE

Same artefact everywhere

- Image digest or package version
- What was tested is what ships
- Config from Parameter Store
- One pipeline, many stages

BUILD PER ENVIRONMENT

Rebuild for each stage

- Dependencies may differ
- Prod runs untested bits
- Config baked into builds
- Often one pipeline per account

THE TIEBREAKER

Production must run exactly what was tested: build once and promote the artefact.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Access denied reading artefacts in target account	Customer managed KMS key and policies
Monorepo, run only the affected service	Trigger file path filters
Only during the change window	Deployment window rule
Alarm after release, return to last good	On-success condition, rollback
What was tested differs from prod	Build once, promote
Rapid commits, deploy only the latest	Superseded execution mode

ON THE EXAM

Find the word that names the failure, then the pipeline feature that removes it.

> Tallis Insurance runs CodePipeline in a tooling account. The artefact bucket uses default SSE-KMS with the AWS managed key.
>
> A new CloudFormation deploy action into the production account fails with access denied when reading the artefact.

What should the solutions architect do?

A. Make the artefact bucket readable by the production account without encryption

B. Re-encrypt with a customer managed KMS key whose key policy, and the bucket policy, allow the production deployment role

C. Add a grant for the production account on the AWS managed key

D. Copy artefacts to a bucket in the production account by hand

◆ PAUSE AND DECIDE

What should the solutions architect do?

A **Remove encryption**

Weakens security and is not needed.

B **Customer managed key**

Its policy can allow another account; bucket policy grants read.

C **Grant on AWS managed key**

AWS managed key policies cannot be changed for other accounts.

D **Manual copy**

Breaks automation.

THE SIGNAL

Cross-account artefacts: customer managed KMS key shared through its key policy.

> Tallis Insurance keeps six microservices in one GitHub monorepo, each in its own folder with its own pipeline. Every push currently starts all six pipelines.
>
> The team wants each pipeline to start only when its own folder changes.

What should the solutions architect configure?

A Split the monorepo into six repositories

B A Lambda function that inspects each webhook and starts the right pipeline

C Polling every five minutes

D V2 pipeline triggers with file path include filters for each service folder

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Split repo

Large change to the team's workflow.

B Lambda dispatcher

Custom code for a built-in feature.

C Polling

Still starts on any change, and slower.

D Path filters

Native trigger filtering per folder.

THE SIGNAL

Monorepo, start only the affected pipeline: trigger file path filters.

> Tallis Insurance policy allows production deployments only between 09:00 and 17:00, Tuesday to Thursday. Pipelines currently run at any hour.
>
> Releases that arrive outside the window should wait automatically, without an engineer approving each one.

What should the solutions architect implement?

A. An entry condition on the production stage with a deployment window rule

B. A manual approval action before production

C. EventBridge schedules that disable and enable the stage transition

D. A change freeze announced by email

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Deployment window rule
The stage waits until the window opens.

B Manual approval
Needs a person for each release.

C Scheduled transitions
Works, but is custom scheduling glue.

D Email freeze
Not enforced.

THE SIGNAL

Deploy only in a time window, without people: entry condition with a window rule.

> Tallis Insurance deploys a CloudFormation stack in its production stage. Twice, a release passed tests but raised the error rate alarm within 20 minutes.
>
> Each time, engineers spent an hour redeploying the previous version. The pipeline should do this automatically.

What should the solutions architect configure?

A Switch the pipeline to parallel execution mode

B A runbook for the on-call engineer

C An on-success condition on the production stage with a CloudWatch alarm rule and a rollback result

D Longer integration tests

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Parallel mode

Stage rollback is not available in parallel mode.

B Runbook

Still manual.

C On-success alarm rule, rollback

Watches the alarm, then redeploys the last good run.

D Longer tests

Helpful, but these faults appear only in production.

THE SIGNAL

Alarm after a stage, return to the last good release: on-success condition with rollback.

> Tallis Insurance has one pipeline per environment account. Each builds a container image from the main branch. A production incident was traced to a library version that differed from the one tested in staging.
>
> The team wants production to run exactly what passed staging.

What should the solutions architect recommend?

A Pin more library versions

B Run tests again in the production pipeline

C Rebuild production only on weekends

D One pipeline in a tooling account that builds the image once and promotes the same digest to each account

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Pin versions

Reduces drift but still rebuilds.

B Retest in production

Tests a different build again.

C Weekend rebuilds

Changes timing, not the cause.

D Build once, promote digest

The tested artefact is the shipped artefact.

THE SIGNAL

Production must match what was tested: build once and promote by digest.

> A new Tallis Insurance subsidiary, with a brand new AWS account, needs Git hosting that stays inside AWS, uses IAM for access, and needs no third-party contract.
>
> An engineer says CodeCommit is closed to new customers, so they must use GitHub.

What should the solutions architect recommend?

A GitHub through CodeConnections

B AWS CodeCommit, which is generally available to new customers again

C Git repositories on an EC2 instance

D Store source in S3 as zip files

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A GitHub

Third party; fails the stated requirement.

B CodeCommit

Managed, IAM-based, and open to new accounts since Nov 2025.

C Git on EC2

Self-managed undifferentiated work.

D S3 zips

Not version control.

THE SIGNAL

AWS-native managed Git with IAM: CodeCommit, open to new customers again.

What to carry out of this module

- **Tooling account** pipeline, deploy roles in target accounts.
- **Customer managed KMS key** for cross-account artefacts.
- **Build once, promote** the same artefact by digest or version.
- **V2 controls**: path filters, window rules, alarm rollback.
- **CodeCommit** is open to new customers again.

> ON THE EXAM
>
> Where the pipeline lives, how it reaches accounts, what gates a release.